

Arjuna JEE 2025

Maths

Trigonometric Identities

- Q1** In a circle of diameter 40 cm the length of a chord is 20 cm. The length of minor arc corresponding to the chord, is equal to
- (A) $\frac{10\pi}{3}$ cm
(B) $\frac{20\pi}{3}$ cm
(C) $\frac{5\pi}{3}$
(D) $\frac{30\pi}{3}$ cm
- Q2** If the angles of a triangle are in ratio 3 : 4 : 5, then smallest angle in degrees and the greatest angle in radians, are
- (A) $50^\circ, \frac{\pi}{12}$
(B) $57^\circ, \frac{\pi}{12}$
(C) $45^\circ, \frac{5\pi}{12}$
(D) $85^\circ, \frac{\pi}{12}$
- Q3** The real number of x and y satisfy the equation $xy = \sin(2t)$ and $\frac{x}{y} = \tan(t)$ where $0 < t < \frac{\pi}{2}$. The value of $x^2 + y^2$, is
- (A) $\sqrt{2}$ (B) 1
(C) 2 (D) 4
- Q4** Exact value of $\frac{\cos 21^\circ - \sin 21^\circ}{\sin^2 57^\circ - \sin^2 33^\circ}$ is equal to
- (A) 2
(B) $-\sqrt{2}$
(C) 1
(D) $\sqrt{2}$
- Q5** Exact value of $\sec 10^\circ - \tan 10^\circ - \tan 40^\circ$ is equal to
- (A) $\sqrt{3}$
(B) 2
(C) 0
(D) $\frac{-1}{\sqrt{3}}$
- Q6** If A and B are complimentary angles, then
- (A) $(1 + \tan \frac{A}{2})(1 + \tan \frac{B}{2}) = 2$
(B) $(1 + \cot \frac{A}{2})(1 + \cot \frac{B}{2}) = 2$
(C) $(1 + \sec \frac{A}{2})(1 + \operatorname{cosec} \frac{B}{2}) = 2$
(D) $(1 - \tan \frac{A}{2})(1 - \tan \frac{B}{2}) = 2$
- Q7** If $A = \tan 6^\circ \tan 42^\circ$ and $B = \cot 66^\circ \cot 78^\circ$ then
- (A) $A = 2B$
(B) $A = \frac{1}{3}B$
(C) $A = B$
(D) $3A = 2B$
- Q8** The value of expression $\sum_{\theta=0}^8 \frac{1}{1+\tan^2(10\theta)^\circ}$ is equal to
- (A) 5
(B) $\frac{21}{4}$
(C) $\frac{14}{3}$
(D) $\frac{9}{2}$
- Q9** If $\tan \alpha, \tan \beta$ are the roots of the equation $x^2 + px + 1 = 0$, then the value of $\sin^2(\alpha + \beta) + p \sin(\alpha + \beta) + \cos(\alpha + \beta)$ is $+ q \cos^2(\alpha + \beta)$
- (A) independent of p but dependent on q
(B) independent of q but dependent on p
(C) independent of both p and q
(D) dependent on both p and q
- Q10**



If $\cos 5\theta = a \cos \theta + b \cos^3 \theta + c \cos^5 \theta + d$, then

- (A) $a = 20$
 (B) $b = -30$
 (C) $a + b + c = 2$
 (D) $a + b + c + d = 1$

Q11 The minimum value of $\frac{\sec^4 \alpha}{\tan^2 \beta} + \frac{\sec^4 \beta}{\tan^2 \alpha}$, $\alpha, \beta \neq \frac{n\pi}{2}$, $n \in \mathbb{Z}$ is

- (A) 4 (B) 8
 (C) 1 (D) 16

Q12 $\sqrt{3} \sin \frac{\pi}{9} (4 + \sec \frac{\pi}{9})$ is equal to

- (A) 3
 (B) $\frac{1}{3}$
 (C) -3
 (D) $-\frac{1}{3}$

Q13 If $\lambda \tan^2 x + \frac{\cot^2 x}{\lambda} = 2$ where $\lambda > 0$ & $x \in (0, \frac{\pi}{2})$ then

$\sin^2 x + 5 \cos^2 x + 3 \sin x \cos x$ is equal to

- (A) $\frac{1}{(\lambda+1)}$
 (B) $\frac{(1-4\lambda)}{(\lambda+1)}$
 (C) $\frac{1-4\lambda+3\sqrt{\lambda}}{(\lambda+1)}$
 (D) $\frac{1+5\lambda+3\sqrt{\lambda}}{(\lambda+1)}$

Q14 $\tan 3^\circ + 2 \tan 6^\circ + 4 \tan 12^\circ + 8 \cot 24^\circ = \cot \theta^\circ$ (0

$< \theta < 90^\circ$), then

- (A) $\cot(10\theta)^\circ = 1$
 (B) $\cot(15\theta)^\circ = 1$
 (C) $\cot \theta^\circ = 0$
 (D) $\cot(15\theta)^\circ = \sqrt{3}$

Q15 If $\sin x + \sin^2 x = 1$, then the value of $\cos^2 x + \cos^4 x$ is

- (A) 0 (B) 2
 (C) 1 (D) None of these

Q16 Consider the statements about $y = \frac{\sin 3x}{\sin x}$, $\sin x \neq 0$ Which of the following is (are) correct?

- (A) The minimum value of y is (-1)
 (B) The maximum value of y is 3
 (C) The minimum value of y is NOT defined
 (D) The maximum value of y is NOT defined

Q17 If $\tan \theta + \cot(30^\circ - \theta) + \cot(150^\circ - \theta)$, then $= k \tan(a\theta)$

- (A) $k = 3$
 (B) $a = 3$
 (C) $k = 1$
 (D) $a = 1$

Q18 $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ and $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$. Based on this information, answer the following questions.

$\sin A + 2 \sin 3A + \sin 5A$ is equal to

- (A) $4 \sin 3A \cos^2 A$
 (B) $4 \cos 3A \sin^2 A$
 (C) $4 \sin 3A \sin^2 A$
 (D) $\cos 3A \cos^2 A$

Q19 $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ and $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$.

Based on this information, answer the following questions. $\frac{\sin A + \sin 2A + \sin 4A + \sin 5A}{\cos A + \cos 2A + \cos 4A + \cos 5A}$ is equal to

- (A) $\cot 3A$
 (B) $\tan 3A$
 (C) $2 \cot 3A$
 (D) $2 \tan 3A$

Q20 $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ and $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$.

Based on this information, answer the following questions. $\frac{\cot 7A + \cot 3A - \cot 5A - \cot A}{\sin 7A - \sin 3A - \sin 5A + \sin A}$ is equal to

- (A) $\cot 2A$
 (B) $\tan 2A$



(C) $2 \cot 2 A$ (D) $2 \tan 2 A$

Q21

Match the items of Column-I with those of Column-II.

Column I		Column II	
(A)	Absolute value of $288 \cos \frac{\pi}{31} \cos \frac{2\pi}{31} \cos \frac{4\pi}{31} \cos \frac{8\pi}{31} \cos \frac{16\pi}{31}$ is	(p)	4
(B)	The value of $[\tan 130^\circ - \tan 10^\circ + \sqrt{3} \tan 130^\circ \tan 10^\circ]^2$ is	(q)	2
(C)	If $\theta = \frac{2017\pi}{5}$, then the value of $2(\sin^2 4\theta + \cos^2 \theta)$ is	(r)	3
(D)	Value of expression $\frac{\sin 56^\circ + \sin 154^\circ + \sin 150^\circ}{\sin 28^\circ \sin 77^\circ \sin 75^\circ}$ is	(s)	9

(A) (A) -s; (B) -r; (C) -q; (D) -p

(B) (A) -p; (B) -q; (C) -r; (D) -s

(C) (A) -s; (B) -r; (C) -p; (D) -q

(D) (A) -r; (B) -s; (C) -q; (D) -p

Q22

Match the items of column-I with of column-II.

Column I		Column II	
(A)	Value of $(\cos 24^\circ \cos 36^\circ \cos 84^\circ)$ is $\frac{\sqrt{a}-1}{b}$ (a and b are coprime numbers), then the value of $(a+b)$ is	(p)	12
(B)	Let $\alpha + \beta = \frac{\pi}{5}$, $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$, if then the maximum value of $\sin \alpha + \sin \beta$ is $\frac{\sqrt{a}-1}{b}$, (a and b are coprime numbers), then the value of $(2a+b)$ is	(q)	21
(C)	Value of $\left(\cot \frac{\pi}{12} + \cot \frac{\pi}{8} - \tan \frac{\pi}{8}\right)$ is $(a + \sqrt{b})$ (a and b are coprime numbers), then the value of $(2a+3b)$ is	(r)	7
(D)	If the solution of $\cos^2 5^\circ + \cos^2 10^\circ + \cos^2 15^\circ + \dots + \cos^2 85^\circ = \frac{a}{b}$ (where a and b are prime numbers), then the value of $(a-5b)$ is	(s)	17

(A) (A) -p; (B) -q; (C) -r; (D) -s

(B) (A) -q; (B) -r; (C) -p; (D) -s

(C) (A) -q; (B) -p; (C) -r; (D) -s

(D) (A) -q; (B) -p; (C) -s; (D) -r

Q23

Match the items of column I with those of column II

Column I		Column II	
(A)	$\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$ is equal to	(p)	0
(B)	If $\tan A + \tan B + \tan C = 6$ and $\tan A \tan B = 2$ where $A + B + C = 180^\circ$, then $\tan C$ is	(q)	$\sqrt{3}$
(C)	The value of $\cot 70^\circ + 4 \cos 70^\circ$ is	(r)	4
(D)	If $\tan^2 \theta = 2 \tan^2 \alpha + 1$, then the value of $\cos 2\theta + \sin^2 \alpha$ is	(s)	3



- (A) (A) -s; (B) -r; (C) -q; (D) -p
 (B) (A) -r; (B) -s; (C) -p; (D) -q
 (C) (A) -r; (B) -s; (C) -q; (D) -p
 (D) (A) -s; (B) -q; (C) -r; (D) -p

Q24 If _____ the _____ expression, $2 \cos 10^\circ + \sin 100^\circ + \sin 1000^\circ + \sin 10000^\circ$ is simplified, then it simplifies to a $\cos$ where a is natural number and b° is acute angle, then $b - a$ is

Q25 The number of integral values of n so that $\sin x(\sin x + \cos x) = n$ has at least one solution, is

Q26 If $\alpha, -\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ is the solution of $4 \cos \alpha + 5 \sin \alpha = 1$, then the value of $\tan \alpha$ is

- (A) $\frac{10-\sqrt{10}}{12}$
 (B) $\frac{\sqrt{10}-10}{12}$
 (C) $\frac{\sqrt{10}-10}{6}$
 (D) $\frac{10-\sqrt{10}}{6}$

Q27 $2 \sin\left(\frac{\pi}{22}\right) \sin\left(\frac{3\pi}{22}\right) \sin\left(\frac{5\pi}{22}\right) \sin\left(\frac{7\pi}{22}\right) \sin\left(\frac{9\pi}{22}\right)$ is equal to

- (A) $\frac{3}{16}$
 (B) $\frac{1}{16}$
 (C) $\frac{1}{32}$
 (D) $\frac{9}{32}$

Q28

The value of $\cot \frac{\pi}{24}$ is

- (A) $\sqrt{2} + \sqrt{3} + 2 - \sqrt{6}$
 (B) $\sqrt{2} + \sqrt{3} + 2 + \sqrt{6}$
 (C) $\sqrt{2} - \sqrt{3} - 2 + \sqrt{6}$
 (D) $3\sqrt{2} - \sqrt{3} - \sqrt{6}$

Q29 The _____ value _____ of

$\cos \frac{\pi}{2^2} \cdot \cos \frac{\pi}{2^3} \cdot \dots \cdot \cos \frac{\pi}{2^{10}} \cdot \sin \frac{\pi}{2^{10}}$ is :

- (A) $\frac{1}{256}$

- (B) $\frac{1}{2}$
 (C) $\frac{1}{512}$
 (D) $\frac{1}{1024}$

Q30 The value of $\sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ$ is

- (A) $\frac{1}{36}$
 (B) $\frac{1}{16}$
 (C) $\frac{1}{32}$
 (D) $\frac{1}{18}$

Q31 If $L = \sin^2\left(\frac{\pi}{16}\right) - \sin^2\left(\frac{\pi}{8}\right)$ and $M = \cos^2\left(\frac{\pi}{16}\right) - \sin^2\left(\frac{\pi}{8}\right)$, then

- (A) $L = -\frac{1}{2\sqrt{2}} + \frac{1}{2}\cos \frac{\pi}{8}$
 (B) $M = \frac{1}{2\sqrt{2}} + \frac{1}{2}\cos \frac{\pi}{8}$
 (C) $M = \frac{1}{4\sqrt{2}} + \frac{1}{4}\cos \frac{\pi}{8}$
 (D) $L = \frac{1}{4\sqrt{2}} - \frac{1}{4}\cos \frac{\pi}{8}$

Q32 If $\tan A = \frac{1}{\sqrt{x(x^2+x+1)}}$, $\tan B = \frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ and

$\tan C = (x^{-3} + x^{-2} + x^{-1})^{1/2}$, $0 < A, B, C$, $< \frac{\pi}{2}$

then $A + B$ is equal to :

- (A) C
 (B) $\frac{\pi}{2} - C$
 (C) $2\pi - C$
 (D) $\pi - C$

Q33 The _____ value _____ of

$2 \sin\left(\frac{\pi}{8}\right) \sin\left(\frac{2\pi}{8}\right) \sin\left(\frac{3\pi}{8}\right) \sin\left(\frac{5\pi}{8}\right) \sin\left(\frac{6\pi}{8}\right) \sin\left(\frac{7\pi}{8}\right)$ is

- (A) $\frac{1}{4\sqrt{2}}$
 (B) $\frac{1}{4}$
 (C) $\frac{1}{8}$
 (D) $\frac{1}{8\sqrt{2}}$

Q34 If $15 \sin^4 \alpha + 10 \cos^4 \alpha = 6$, for some $\alpha \in R$, then the value of $27 \sec^6 \alpha + 8 \operatorname{cosec}^6 \alpha$ is equal to:

- (A) 350 (B) 500



(C) 400

(D) 250

Q35 If $\cos(\alpha + \beta) = \frac{3}{5}$, $\sin(\alpha - \beta) = \frac{5}{13}$ and $0 < \alpha, \beta < \frac{\pi}{4}$, then $\tan(2\alpha)$ is equal to :

- (A) $\frac{21}{16}$
 (B) $\frac{63}{52}$
 (C) $\frac{33}{52}$
 (D) $\frac{63}{16}$

Q36 The value of $36(4\cos^2 9^\circ - 1)(4\cos^2 27^\circ - 1)(4\cos^2 81^\circ - 1)(4\cos^2 243^\circ - 1)$ is

- (A) 54 (B) 18
 (C) 27 (D) 36

Q37 $96 \cos \frac{\pi}{33} \cos \frac{2\pi}{33} \cos \frac{4\pi}{33} \cos \frac{8\pi}{33} \cos \frac{16\pi}{33}$ is equal to

- (A) 3 (B) 2
 (C) 4 (D) 1

Q38 If $0 < x, y < \pi$ and $\cos x + \cos y - \cos(x + y) = 3/2$, Then, $\sin x + \cos y$ is equal to:

- (A) $\frac{1+\sqrt{3}}{2}$
 (B) $\frac{1-\sqrt{3}}{2}$
 (C) $\frac{\sqrt{3}}{2}$
 (D) $\frac{1}{2}$

Q39 The set of all values of λ for which the equation $\cos^2 2x - 2\sin^4 x - 2\cos^2 x = \lambda$

- (A) $[-2, -1]$
 (B) $[-2, -\frac{3}{2}]$
 (C) $[-1, -\frac{1}{2}]$
 (D) $[-\frac{3}{2}, -1]$

Q40 If $\tan 15^\circ + \frac{1}{\tan 75^\circ} + \frac{1}{\tan 105^\circ} + \tan 195^\circ = 2a$, then the value of $(a + \frac{1}{a})$ is:

- (A) 4
 (B) $4 - 2\sqrt{3}$

(C) 2

(D) $5 - \frac{3}{2}\sqrt{3}$

Q41 The value of $\cos \frac{\pi}{65} \cos \frac{2\pi}{65} \cos \frac{4\pi}{65} \cos \frac{8\pi}{65} \cos \frac{16\pi}{65} \cos \frac{32\pi}{65}$ is

- (A) $\frac{1}{8}$
 (B) $\frac{1}{16}$
 (C) $\frac{1}{32}$
 (D) none of these

Q42 The value of $\sin^2(\frac{\pi}{18}) + \sin^2(\frac{\pi}{9}) + \sin^2(\frac{7\pi}{18}) + \sin^2(\frac{4\pi}{9})$ is

- (A) 1 (B) 2
 (C) 4 (D) none of these

Q43 The value of $\frac{\cot x - \tan x}{\cot 2x}$ is

Q44 The value of $\frac{2(\sin 2x + 2\cos^2 x - 1)}{\cos x - \sin x - \cos 3x + \sin 3x}$ is

- (A) $\cos x$
 (B) $\sec x$
 (C) $\operatorname{cosec} x$
 (D) $\sin x$

Q45 The value of $2\sin^2 B + 4\cos(A+B)\sin A \sin B + \cos$ is

- $2(A+B)$
 (A) 0 (B) $\cos 3A$
 (C) $\cos 2A$ (D) none of these

Q46 If α and β are acute angles satisfying $\cos 2\alpha = \frac{3\cos 2\beta - 1}{3 - \cos 2\beta}$, then $\tan \alpha =$

- (A) $\sqrt{2} \tan \beta$
 (B) $\frac{1}{\sqrt{2}} \tan \beta$
 (C) $\sqrt{2} \cot \beta$
 (D) $\frac{1}{\sqrt{2}} \cot \beta$

Q47 The radian measure corresponding to $-37^\circ 30'$ is equal to

- (A) $-\frac{5\pi}{24}$
 (B) $\frac{\pi}{24}$



- (C) $\frac{5\pi}{24}$
 (D) $\frac{-7\pi}{24}$

Q48 The expression,

$$\frac{\tan\left(\frac{3\pi}{2}-\alpha\right)\cos\left(\frac{3\pi}{2}-\alpha\right)}{\cos(2\pi-\alpha)} + \cos\left(\alpha-\frac{\pi}{2}\right)\sin(\pi-\alpha) + \cos\left(\pi+\alpha\right)\sin\left(\alpha-\frac{\pi}{2}\right), \alpha \neq \frac{n\pi}{2}, n \in I,$$

when simplified reduces to

- (A) 0 (B) 1
 (C) -1 (D) 2

Q49 The expression

$$f(x) = \frac{2\sin^2 x - 1}{\cos x} + \frac{\cos x(1+2\sin x)}{1+\sin x} \quad \text{wherever}$$

defined simplifies to

- (A) $\tan^2 x$
 (B) $\tan x$
 (C) $1 - \cos x$
 (D) $1 + \sin^2 x$

Q50 If $A = 10^\circ$, then the value of expression

$$\left(\frac{\sec 2A}{(1-4\sin^2 2A)} + \frac{\operatorname{cosec} A}{(4\cos^2 A-1)} \right)^2 \text{ is}$$

- (A) 6 (B) 8
 (C) 16 (D) 24



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Answer Key

Q1 (B)

Q2 (C)

Q3 (C)

Q4 (D)

Q5 (C)

Q6 (A)

Q7 (C)

Q8 (A)

Q9 (B)

Q10 (D)

Q11 (B)

Q12 (A)

Q13 (D)

Q14 (B)

Q15 (C)

Q16 (A, D)

Q17 (A, B)

Q18 (A)

Q19 (B)

Q20 (A)

Q21 (A)

Q22 (D)

Q23 (C)

Q24 9

Q25 2

Q26 (B)

Q27 (B)

Q28 (B)

Q29 (C)

Q30 (B)

Q31 (B)

Q32 (A)

Q33 (C)

Q34 (D)

Q35 (D)

Q36 (D)

Q37 (A)

Q38 (A)

Q39 (D)

Q40 (A)

Q41 (D)

Q42 (B)

Q43 2

Q44 (C)

Q45 (C)

Q46 (A)

Q47 (A)

Q48 (A)

Q49 (B)

Q50 (C)

